

DR. DESMOND ADEMILUYI

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SENIOR SOFTWARE ENGINEER | PRODUCT-DRIVEN BUILDER | TECH LEAD

User-focused and pragmatic senior engineer with 10+ years of experience delivering scalable platforms and modern web applications. Proven leader of high-impact projects across SaaS, adtech, marketplace, eCommerce, and media domains. Highly skilled in React/Next.js, TypeScript, Python and API integrations, with a strong focus on performance, usability, and seamless delivery. Known for driving iterative product development, solving complex problems with simplicity, and collaborating effectively across functions.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), HTML5, SCSS, Python, Flutter

Frameworks/Libraries: React.js, Next.js, Tailwind CSS, Redux, GraphQL, Contentful CMS

Backend: Node.js, NestJS, Express.js, Django, FastAPI

Databases: PostgreSQL, Firebase, MongoDB, MySQL

Tools & DevOps: GitHub Actions, Docker, CI/CD pipelines, Vercel, Stripe, Kafka

Other: Recurly, Shopify Liquid, Flutter, BrightScript, Swift/TVMMLKit JS

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Talent Systems — Los Angeles, CA | Jun 2023 – Present

- Developed production-grade, mobile-friendly web apps using Typescript, Next.js, React, GraphQL and CSS.
- Led the implementation of real-time paywall flows using Stripe and Recurly, ensuring secure, seamless user interactions.
- Modernized legacy components into modular, maintainable architecture using TypeScript and GraphQL.
- Promoted strong CI/CD, testing, and code review practices, helping reduce bug reports by 35%.
- Delivered impactful user-facing features with fast turnaround, balancing technical debt and speed.
- Mentored junior engineers; facilitated workshops on React performance and system design.

Software Engineer

Lovevery — Boise, ID | Oct 2020 – May 2023

- Led Next.js migration and complete TypeScript adoption across the customer-facing frontend.
- Delivered high-performance subscription checkout and claims interface with robust API integrations.
- Designed dynamic UI components using Contentful and reusable component libraries.
- Championed lazy loading, image optimization, and code-splitting, improving load times by 20%.

Senior Software Engineer (Contract)

Antifragility Health — Anaheim, CA | Aug 2020 – Oct 2020

- Developed a health-focused insurance checkout and product claims portal with React and Node.js.
- Designed resilient UI components with data validation and user-friendly error handling.

- Integrated secure payment processing and modular claims status tracking.

Full-Stack Developer

PilotX — Laguna Beach, CA | Sep 2017 – Aug 2020

- Built and launched a new Ad Tech platform for connecting advertisers and publishers using React and Python.
- Built a customizable, real-time video platform with React/Next.js and Node.js.
- Architected backend dashboard and API gateways with an event-driven architecture.
- Used SSR, prefetching, and optimized routes to drive SEO performance gains of 18%.

LEADERSHIP & MENTORSHIP

- Led engineering mentorship programs at Talent Systems and PilotX.
- Adjunct Instructor, Westcliff University (2023–Present): Teaching graduate and undergraduate courses in software engineering, systems design, and DevOps practices.

EDUCATION

Doctor of Science in Computer Science

Aspen University — Phoenix, AZ — 2023

M.Tech. & B.Tech. in Computer Science

Ladoke Akintola University — Nigeria — 2015, 2010

Coding Bootcamp Certificate

University of California, Irvine — 2017

SELECTED PROJECTS

Pregnants.com

Non-invasive pregnancy detection web app using PPG signal processing and machine learning.

- Built with Next.js, NestJS, and Python, with real-time video analysis and signal feature extraction.
- Designed an intuitive user testing experience with responsive UI and live feedback.

Arewethere.yt

Live countdown app for special events.

- Built with React.js and Node.js.
- Supports real-time updates, custom sharing, and persistent browser sessions.

PATENTS

Intelligent Conversational Video (ICV) System (Patent Pending)

USPTO Application No: 63/780,221 | Filed: March 2025

PUBLICATIONS & RESEARCH

- "Deepfake Video Detection Using Human Facial Features" (Doctoral Dissertation, 2023)
- "Unraveling the Impact of React.memo in Functional Components" (Medium Article)
- Peer-reviewed papers on biometric authentication and applied AI in real-time media systems